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Q1- Express the change in internal energy of a system when
(i) No heat is absorbed by the system from the surroundings, but work (w) is done on the system. What type of wall does the system have ?
(ii) No work is done on the system, but q amount of heat is taken out from the system and given to the surroundings. What type of wall does the system have?
(iii) w amount of work is done by the system and q amount of heat is supplied to the system. What type of system would it be?

Q2- Two litres of an ideal gas at a pressure of 10 atm expands isothermally at 25 °C into a vacuum until its total volume is 10 litres. How much heat is absorbed and how much work is done in the expansion ?

Q3- Consider the same expansion, but this time against a constant external pressure of 1 atm.

Q4- Consider the expansion given in problem 5.2, for 1 mol of an ideal gas conducted reversibly.

Q5- If water vapour is assumed to be a perfect gas, molar enthalpy change for vapourisation of 1 mol of water at 1bar and 100°C is 41kJ mol–1. Calculate the internal energy change, when 1 mol of water is vapourised at 1 bar pressure and 100°C.

Q6- 1g of graphite is burnt in a bomb calorimeter in excess of oxygen at 298 K and 1 atmospheric pressure according to the equation C (graphite) + O_2 (g) → CO_2 (g) During the reaction, temperature rises from 298 K to 299 K. If the heat capacity of the bomb calorimeter is 20.7kJ/K, what is the enthalpy change for the above reaction at 298 K and 1 atm?

Q7- A swimmer coming out from a pool is covered with a film of water weighing about 18g. How much heat must be supplied to evaporate this water at 298 K ? Calculate the internal energy of vaporisation at 298K. $\Delta_{vap}H^0$ for water at 298K= 44.01kJ mol^{-1}

Q8- The combustion of one mole of benzene takes place at 298 K and 1 atm. After combustion, CO2(g) and H_2O (1) are produced and 3267.0 kJ of heat is liberated. Calculate the standard enthalpy of formation, Δf H0 of benzene. Standard enthalpies of formation of CO_2(g) and H_2O(l) are –393.5 kJ mol^{-1} and – 285.83 kJ mol^{-1} respectively.

Q9- Predict in which of the following, entropy increases/decreases :
(i) A liquid crystallizes into a solid.
(ii) Temperature of a crystalline solid is raised from 0 K to 115 K.
(iii) $2NaHCO_3$ (s) → Na_2CO_3 (s) + CO_2 (g) + H_2O(g)
(iv) H_2(g) → 2H(g)

Q10- For oxidation of iron,
4Fe(s)+30, (g) → $2Fe_2O_3$ (s)
entropy change is 549.4 JK mol - at 298 K. Inspite of negative entropy change of this reaction, why is the reaction spontaneous? (ΔH for this reaction is -1648 x 10^3 J mol^{-1})

Q11- Calculate Δ_rG^0 for conversion of oxygen to ozone, 3/2 O_2(g) → O_3(g) at 298 K. if Kp for this conversion is 2.47×10^{-29}.

Q12- Find out the value of equilibrium constant for the following reaction at 298 K.
$2NH_3$ (g) + CO_2 (g) ⇌ NH_2CONH_2 (aq)+ H_2O (l) Standard Gibbs energy change, ΔG at the given temperature is -13.6 kJ mol^{-1}.

Q13- At 60°C, dinitrogen tetroxide is 50 per cent dissociated. Calculate the standard free energy change at this temperature and at one atmosphere.

Q14- Choose the correct answer. A thermodynamic state function is a quantity
(i) used to determine heat changes
(ii) whose value is independent of path
(iii) used to determine pressure volume work
(iv) whose value depends on temperature only.

Q15- For the process to occur under adiabatic conditions, the correct condition is:
(i) $\Delta T = 0$
(ii) $\Delta p = 0$
(iii) $q = 0$
(iv) $w = 0$

Q16- The enthalpies of all elements in their standard states are:
(i) unity
(ii) zero
(iii) < 0
(iv) different for each element

Q17- $\Delta U\,0$ of combustion of methane is – X kJ mol^{-1}. The value of $\Delta H0$ is
(i) = $\Delta U\,0$
(ii) > $\Delta U\,0$
(iii) < $\Delta U\,0$
(iv) = 0

Q18- The enthalpy of combustion of methane, graphite and dihydrogen at 298 K are, –890.3 kJ mol^{-1} ,–393.5 kJ mol^{-1}, and –285.8 kJ mol^{-1} respectively. Enthalpy of formation of $CH_4(g)$ will be
(i) –74.8 kJ mol^{-1}
(ii) –52.27 kJ mol^{-1}
(iii) +74.8 kJ mol^{-1}
(iv) +52.26 kJ mol^{-1}.

Q19- A reaction, A + B → C + D + q is found to have a positive entropy change. The reaction will be
(i) possible at high temperature
(ii) possible only at low temperature
(iii) not possible at any temperature
(v) possible at any temperature

Q20- In a process, 701 J of heat is absorbed by a system and 394 J of work is done by the system. What is the change in internal energy for the process?

Q21- The reaction of cyanamide, NH_2CN (s), with dioxygen was carried out in a bomb calorimeter, and ΔU was found to be –742.7 kJ mol^{-1} at 298 K. Calculate enthalpy change for the reaction at 298 K.

$$NH_2CN(g) + \frac{3}{2}O_2(g) \rightarrow N_2(g) + CO_2(g) + H_2O(l)$$

Q22- Calculate the number of kJ of heat necessary to raise the temperature of 60.0 g of aluminium from 35°C to 55°C. Molar heat capacity of Al is 24 J mol^{-1} K^{-1}.

Q23- Calculate the enthalpy change on freezing of 1.0 mol of water at10.0°C to ice at –10.0°C. $\Delta_{fus}H$ = 6.03 kJ mol^{-1} at 0°C. Cp [$H_2O(l)$] = 75.3 J mol^{-1} K^{-1} Cp [$H_2O(s)$] = 36.8 J mol^{-1} K^{-1}

Q24- Enthalpy of combustion of carbon to CO_2 is –393.5 kJ mol^{-1}. Calculate the heat released upon formation of 35.2 g of CO_2 from carbon and dioxygen gas.

Q25- Enthalpies of formation of CO(g), $CO_2(g)$, $N_2O(g)$ and $N_2O_4(g)$ are –110, – 393, 81 and 9.7 kJ mol^{-1} respectively. Find the value of Δ_rH for the reaction: $N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)$

Q26- Given $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$; Δ_rH^0 = –92.4 kJ mol^{-1} What is the standard enthalpy of formation of NH_3 gas?

Q27- Calculate the standard enthalpy of formation of CH3OH(l) from the following data:

$CH_3OH\ (l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$; Δ_rH^0 = –726 kJ mol^{-1}

C(graphite) + $O_2(g) \rightarrow CO_2(g)$; Δ_cH^0 = –393 kJ mol^{-1}

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$; $\Delta_f H^0$ = –286 kJ mol^{-1}.

Q28- Calculate the enthalpy change for the process $CCl_4(g) \rightarrow C(g) + 4\ Cl(g)$ and calculate bond enthalpy of C – Cl in $CCl_4(g)$. $\Delta_{vap}H^0(CCl_4)$ = 30.5 kJ mol^{-1}. $\Delta_f H_0$ (CCl_4) = –135.5 kJ mol^{-1}. Δ_aH^0 (C) = 715.0 kJ mol^{-1}, where Δ_aH^0 is enthalpy of atomisation Δ_aH^0 (Cl_2) = 242 kJ mol^{-1}

Q29- For an isolated system, $\Delta U = 0$, what will be ΔS ?

Q30- For the reaction at 298 K, 2A + B → C ΔH = 400 kJ mol^{-1} and ΔS = 0.2 kJ K–1 mol^{-1} At what temperature will the reaction become spontaneous considering ΔH and ΔS to be constant over the temperature range.

Q31- For the reaction, $2Cl(g) \rightarrow Cl_2(g)$, what are the signs of ΔH and ΔS ?

Q32- For the reaction 2A(g) + B(g) → 2D(g) ΔU^0 = –10.5 kJ and ΔS^0 = –44.1 JK^{-1}. Calculate ΔG^0 for the reaction, and predict whether the reaction may occur spontaneously.

Q33- The equilibrium constant for a reaction is 10. What will be the value of $\Delta G0$? R = 8.314 JK^{-1} mol^{-1}, T = 300 K.

Q34- Comment on the thermodynamic stability of NO(g), given

$\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightarrow NO(g)$; Δ_rH^0 = 90 kJ mol^{-1}

$NO(g) + \frac{1}{2}O_2(g) \rightarrow NO_2(g)$: Δ_rH^0 = –74 kJ mol^{-1}

Q35- Calculate the entropy change in surroundings when 1.00 mol of $H_2O(l)$ is formed under standard conditions. $\Delta_f H^0$ = –286 kJ mol^{-1}.